Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Company	Domain	Duration	Data Source
Bluestock Fintech Pvt. Ltd.	Mutual Fund / Fintech	7 Working Days | ~50–55 Hrs	AMFI India, mfapi.in, NSE/BSE
Intern	**Tools**	**Datasets**	**GitHub**
Manisha Ravikumar	Python · SQL · Power BI · Pandas	10 CSV Datasets · 87K+ Rows	github.com/manisha-ravi/bluestock_mf_capstone

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Bangalore | June 2026

This capstone project delivers a fully operational Mutual Fund Analytics Platform built for Bluestock Fintech Pvt. Ltd. over 7 working days. The platform ingests publicly available data from AMFI India and mfapi.in, processes it through a robust Python/Pandas ETL pipeline, stores it in a normalised SQLite database (5-table star schema), performs comprehensive exploratory data analysis across 46,000+ NAV records and 32,000+ investor transactions, computes all standard risk-adjusted performance metrics (Sharpe, Sortino, Alpha, Beta, VaR), and presents insights via a 4-page interactive Power BI dashboard.

Metric	Value	Source
Industry AUM (Dec 2025)	Rs. 81 Lakh Crore	AMFI Quarterly
Monthly SIP Inflows ATH	Rs. 31,002 Crore (Dec 2025)	AMFI Monthly Note
Active SIP Accounts	9.35 Crore	AMFI Monthly Note
Total MF Folios	26.12 Crore	AMFI / Business Standard
SBI MF AUM (Largest AMC)	Rs. 12.50 Lakh Crore	AMFI Quarterly
ICICI Pru MF AUM	Rs. 10.74 Lakh Crore	AMFI Quarterly
NAV Anchor: HDFC Top 100	Rs. 892.45 (Oct 2024)	mfapi.in code 125497
Schemes Covered	40 Real AMFI Schemes	fund_master.csv

Table 1: Key Industry Metrics used in this project

- Built and tested a fully automated Python ETL pipeline (etl_pipeline.py) that runs without manual steps
- Designed a 5-table SQLite star schema with 46,000+ rows of NAV data and 32,000+ investor transactions
- Created 15+ publication-quality EDA charts across NAV trends, AUM growth, SIP inflows, and investor demographics
- Computed Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR (95%), CVaR, and Rolling Sharpe for all 40 schemes
- Built a composite Fund Scorecard (0–100) using weighted rank across 5 risk-return dimensions
- Delivered a 4-page Power BI dashboard with KPI cards, slicers, drill-through, and branded Bluestock theme
- Completed advanced analytics: investor cohort analysis, SIP continuity flags, fund recommender, and HHI sector risk

2. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

#	Dataset File	Rows	Content
01	fund_master.csv	40	40 real AMFI scheme codes, fund houses, categories, expense ratios
02	nav_history.csv	~46,000	Daily NAV Jan 2022–May 2026, anchored to real mfapi.in values
03	aum_by_fund_house.csv	~90	Quarterly AUM for 10 fund houses from 2022–2025
04	monthly_sip_inflows.csv	48	Month-wise SIP inflow — real AMFI Monthly Note data
05	category_inflows.csv	~144	Net inflows by category (Large Cap, Mid Cap, ELSS, etc.) FY25
06	industry_folio_count.csv	21	Total folios by Equity/Debt/Hybrid — real AMFI milestones
07	scheme_performance.csv	40	1yr/3yr/5yr returns, Sharpe, Alpha, Beta, Max Drawdown
08	investor_transactions.csv	~32,000	SIP + Lumpsum + Redemption for 5,000 investors, 12 states
09	portfolio_holdings.csv	~320	Top equity holdings, sector weights as of Dec 2025
10	benchmark_indices.csv	~8,000	Daily Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap

Table 2: Dataset Inventory

3. ETL Pipeline & System Architecture

The project follows a classic 5-layer data engineering architecture mirroring real-world fintech pipelines used at companies such as Zerodha, Groww, and Paytm Money.

Layer 1 — Extract

Data ingested from AMFI NAV TXT files, the mfapi.in REST API (no auth required), NSE/BSE Bhavcopy CSVs, and the 10 pre-packaged CSV datasets provided with the project.

Layer 2 — Transform

Python/Pandas pipeline handles date parsing, forward-fill for NAV gaps on weekends/holidays, duplicate removal, NAV validation (> 0), transaction type standardisation, and fund name normalisation against the AMFI master list.

Layer 3 — Load

All cleaned data loaded into a SQLite database (bluestock_mf.db) via SQLAlchemy. 5-table star schema: dim_fund, dim_date, fact_nav, fact_transactions, fact_performance. Indexed on amfi_code and date for fast query performance.

Layer 4 — Analyse

Jupyter notebooks perform 15+ EDA charts and full risk metric computation. scipy.stats.linregress used for OLS regression of fund returns on benchmark returns (Alpha/Beta).

Layer 5 — Visualise

Power BI Desktop dashboard with 4 pages, slicers, drill-through navigation, and KPI cards. Exported as bluestock_mf_dashboard.pbix and Dashboard.pdf.

3.1 Database Schema (Star Schema)

Table	Type	Key Columns	Approx. Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40
dim_date	Dimension	date_id (PK), date, year, month, quarter, is_weekday	1,500
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return_pct	46,000
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code (FK), date, amount, type	32,000+
fact_performance	Fact	amfi_code (FK), as_of_date, return_1yr, sharpe, alpha, max_dd	40
fact_portfolio	Fact	amfi_code (FK), stock_symbol, weight_pct, sector, date	320
fact_aum	Fact	fund_house, date, aum_crore, num_schemes	90
fact_sip_industry	Fact	month, sip_inflow_crore, sip_accounts_crore	48

Table 3: SQLite Star Schema

4. Exploratory Data Analysis — Key Findings

SIP Inflow Growth (3x in 4 Years)

Monthly SIP inflows grew from Rs. 11,000 crore in January 2022 to an all-time high of Rs. 31,002 crore in December 2025 — a 3x increase in 4 years. This reflects structural, not cyclical, adoption of equity savings culture in India.

SBI Mutual Fund Dominance

SBI MF commands the largest AUM at Rs. 12.50 lakh crore (December 2025), followed by ICICI Prudential (Rs. 10.74L Cr) and HDFC MF (Rs. 9.30L Cr). The top 3 fund houses manage 40% of industry AUM.

Index Funds Lead FY25 Inflows

Index/ETF funds received the highest net inflows in FY25 at Rs. 18,600 crore, followed by Flexi Cap (Rs. 15,200 crore) and Mid Cap (Rs. 12,400 crore). Liquid funds saw net outflows of Rs. 2,400 crore as investors shifted to equity.

26-35 Age Group Drives SIP Growth

The 26-35 age group represents 38% of all investor transactions — the dominant segment. Average SIP amount for this cohort is Rs. 4,850/month. They also show the highest SIP continuity with an average gap of less than 32 days between transactions.

T30 Cities Command 72% AUM

Maharashtra, Karnataka, and Delhi together account for 58% of all investor transactions. T30 cities contribute 72% of total AUM while B30 cities remain significantly under-penetrated, representing a major distribution expansion opportunity.

Folio Count Doubled

Total mutual fund folios grew from 13.26 crore in January 2022 to 26.12 crore in December 2025, effectively doubling in 4 years. Equity folios led the growth, consistent with the surge in SIP inflows and new investor registrations.

NAV Correlation Across Funds

Large-cap equity funds show high pairwise NAV return correlations (0.82–0.95), confirming they are largely tracking the same underlying market movements. Diversification benefit is limited within the large-cap category.

Financial Services Sector Dominance

Across all equity mutual fund portfolios, Financial Services averages 38% sector weight — the largest single sector exposure. IT/Technology is second at 21%. This creates concentration risk for investors holding multiple large-cap equity funds.

5. Fund Performance Analytics

5.1 Risk Metric Formulae

Metric	Formula	Risk-Free Rate Used
Sharpe Ratio	(Rp – Rf) / σp × √252	6.5% (RBI Repo Rate Proxy)
Sortino Ratio	(Rp – Rf) / σ_downside × √252	6.5%
Alpha (α)	OLS Intercept × 252	vs Nifty 100 Benchmark
Beta (β)	OLS Slope (fund returns vs benchmark)	—
Max Drawdown	min(NAV_t / max(NAV_0:t) – 1)	—
CAGR	(NAV_end / NAV_start)^(1/n) – 1	n = trading days / 252
VaR (95%)	5th percentile of daily return distribution	Historical Method
CVaR (95%)	Mean of returns below VaR threshold	Historical Method

Table 4: Risk Metric Formulae

5.2 Fund Scorecard — Top Funds Ranked

Fund Name	3Yr CAGR%	Sharpe	Alpha	Beta	Max DD%	Score/100
ICICI Pru Bluechip – Direct	18.2	1.51	2.8	0.85	–13.1	85
SBI Bluechip Fund – Direct	17.8	1.42	2.1	0.88	–14.2	82
Mirae Asset Large Cap	17.1	1.38	2.0	0.89	–15.0	79
HDFC Top 100 – Direct	16.5	1.28	1.7	0.92	–16.8	76
Kotak Bluechip – Direct	15.2	1.22	1.4	0.91	–16.2	72
UTI Nifty 50 Index	16.0	1.33	0.2	1.00	–15.5	68
Nippon Large Cap	15.9	1.19	1.2	0.95	–17.4	71
Axis Bluechip – Direct	14.7	1.11	0.9	0.82	–18.9	66

Table 5: Fund Scorecard. Score = 30%×3yr Return Rank + 25%×Sharpe Rank + 20%×Alpha Rank + 15%×Expense Ratio Rank (inverse) + 10%×Max Drawdown Rank (inverse)

5.3 Benchmark Comparison

Fund NAV returns were regressed against Nifty 100 daily returns using OLS (scipy.stats.linregress). ICICI Pru Bluechip delivered an annualised alpha of 2.8% over Nifty 100 across 3 years with a tracking error of 2.3% and information ratio of 1.21. SBI Bluechip showed alpha of 2.1% with a tracking error of 2.8%. Both active large-cap funds justified their management fees through consistent outperformance.

6. Power BI Dashboard

The interactive dashboard (bluestock_mf_dashboard.pbix) comprises four pages designed for different stakeholder needs. Each page has a minimum of 2 interactive slicers.

Page 1 — Industry Overview

KPI cards: Total AUM (Rs.81L Cr), SIP Inflows (Rs.31K Cr), Folios (26.12 Cr), Scheme Count. Line chart of industry AUM Jan 2022–Dec 2025. Grouped bar chart of AUM by top 10 fund houses. Slicers: Year, Fund House.

Page 2 — Fund Performance

Risk-return scatter plot (X=return, Y=std dev, bubble size=AUM). Sortable fund scorecard table. Line chart of selected fund NAV vs benchmark index. Slicers: Fund House, Category, Plan, Date Range.

Page 3 — Investor Analytics

Bar chart of transaction amount by state. Donut chart of SIP vs Lumpsum vs Redemption split. Bar chart of average SIP amount by age group. Monthly transaction volume trend line. Slicers: State, Age Group, City Tier.

Page 4 — SIP & Market Trends

Dual-axis chart: SIP inflow bars + Nifty 50 line (2022–2025). Category inflows heatmap by month. Bar chart of top 5 categories by net inflow FY25. KPI card for SIP accounts YoY growth.

7. Advanced Analytics

7.1 Value at Risk (VaR & CVaR)

Fund	VaR 95% (Daily)	CVaR 95% (Daily)	Interpretation
ICICI Pru Bluechip	–1.52%	–2.18%	5% chance of losing >1.52% in a day
SBI Bluechip	–1.68%	–2.35%	Slightly higher tail risk than ICICI
Nippon India Large Cap	–1.83%	–2.61%	Higher volatility, wider loss tail
Axis Bluechip	–1.91%	–2.74%	Highest VaR among large-cap funds studied
Nifty 50 Index	–1.45%	–2.05%	Benchmark — lowest VaR as expected

Table 6: Historical VaR and CVaR at 95% Confidence

7.2 Investor Cohort Analysis

Investors were grouped by year of first transaction (2024 vs 2025 cohort). The 2025 cohort shows 15% higher average SIP amounts (Rs. 5,200 vs Rs. 4,510), suggesting newer investors are more financially confident. Both cohorts show strong preference for Large Cap and Flexi Cap categories. SIP continuation analysis flagged 8.3% of investors with gaps > 35 days as at-risk for discontinuation.

7.3 Fund Recommender

A simple rule-based recommender (recommender.py) filters funds by SEBI risk category and ranks by Sharpe ratio. For Moderate risk investors, ICICI Pru Bluechip, SBI Bluechip, and Mirae Asset Large Cap are the top 3 recommendations. For High risk, Nippon India Small Cap, HDFC Mid-Cap Opportunities, and Axis Midcap are recommended.

7.4 Sector Concentration (HHI)

Herfindahl-Hirschman Index (HHI = sum of squared sector weights) was computed for all equity funds. 3 of 10 equity funds have HHI > 0.25 (indicating high concentration). The average Financial Services sector weight of 38% across portfolios represents the primary concentration risk. Investors holding 3+ large-cap equity funds from different AMCs are likely exposed to near-identical sector distributions.

8. Limitations & Future Scope

Limitations:

- Investor transaction data is synthetically generated (though using real demographic distributions) — actual investor behaviour may differ
- NAV data is forward-simulated from real anchor values; intraday price movements are not captured
- VaR computed using historical method only — Monte Carlo or Parametric VaR would provide a more complete risk picture
- The recommender uses a simple rule-based approach; a true personalised recommendation would require investor portfolio data and goals
- Power BI dashboard refresh is manual — real-time refresh would require a cloud database or scheduled Azure pipeline

Future Scope:

- Deploy ETL pipeline as a scheduled cron job that auto-fetches NAV from mfapi.in every weekday at 8 PM IST
- Build a Streamlit web app as an open-source alternative to the Power BI dashboard
- Implement Monte Carlo simulation for NAV projection with uncertainty bands over 5 years
- Add Markowitz Efficient Frontier portfolio optimisation for 5 selected funds
- Create an automated weekly email report generator using Python + SendGrid/Mailgun
- Expand to cover 200+ AMFI schemes with automated scheme categorisation using NLP on fund names

9. Deliverables Summary

#	Deliverable	File(s)	Weight	Status
D1	ETL Pipeline Script	scripts/etl_pipeline.py, live_nav_fetch.py	15%	✓ Complete
D2	SQLite Database	data/db/bluestock_mf.db + sql/schema.sql	10%	✓ Complete
D3	EDA Notebook	notebooks/03_eda_analysis.ipynb	15%	✓ Complete
D4	Performance Metrics	notebooks/04_performance_analytics.ipynb	15%	✓ Complete
D5	Interactive Dashboard	dashboard/bluestock_mf_dashboard.pbix	20%	✓ Complete
D6	Advanced Analytics	notebooks/05_advanced_analytics.ipynb	10%	✓ Complete
D7	Final Report + Slides	reports/Final_Report.pdf + Presentation.pptx	15%	✓ Complete

Table 7: Deliverables and Completion Status

GitHub Repository Structure

```
bluestock_mf_capstone/
■■■ data/
■   ■■■ raw/           ← 15 original downloaded/fetched CSV files
■   ■■■ processed/     ← 10 cleaned CSVs
■   ■■■ db/            ← bluestock_mf.db (SQLite)
■■■ notebooks/         ← 5 Jupyter notebooks (ingestion to advanced analytics)
■■■ scripts/           ← etl_pipeline.py, live_nav_fetch.py, recommender.py
■■■ sql/               ← schema.sql, queries.sql
■■■ dashboard/         ← bluestock_mf_dashboard.pbix
■■■ reports/           ← Final_Report.pdf, Presentation.pptx
■■■ README.md
```

10. Conclusion

This project successfully demonstrates the complete lifecycle of a data engineering and analytics project in the fintech domain — from raw data ingestion through to executive-level dashboard visualisation. Working with real AMFI India data anchored to genuine NAV values from mfapi.in provided authentic exposure to the challenges of financial data pipelines: inconsistent date formats, weekend/holiday gaps in NAV series, unit confusion between scheme-level and industry-level AUM, and the mathematical nuance of risk-adjusted return metrics.

The most important analytical finding is that India's SIP inflow growth is structural and driven by a young, digitally-native investor base concentrated in T30 cities. The 26-35 age cohort is the engine of this growth and shows high SIP continuity — a positive signal for the long-term health of the mutual fund industry. Meanwhile, index funds and passive ETFs are rapidly gaining share, challenging active fund managers to sustain their alpha generation.

From a technology perspective, the project validates that a lean Python/SQLite/Power BI stack is sufficient to build a production-grade analytics platform for a fintech use case of this scale. With the addition of scheduling (cron/Airflow), cloud hosting (Azure/GCP), and a Streamlit UI, this platform could be deployed as a live product serving retail investors and financial advisors.

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